Exploratory Data Analysis & Early Figures

ECON 692 – Applied Economics Seminar

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Today’s Agenda

Note: this is a 2-hour morning session in LM 244B, not our usual evening class.

Time	Duration	Activity
10:00	10 min	Welcome & check-in
10:10	30 min	EDA for causal inference
10:40	20 min	Early figures workshop
11:00	50 min	Work sprint
11:50	10 min	Wrap-up & next week

Final Presentations: Schedule

Two sessions, 20 minutes per student (15-min presentation + 5-min Q&A):

Date	Time	Note
May 7	4:35–5:45	$\sim$4 presentations, then department party
May 14	4:35–8:15	$\sim$9 presentations (full session)

Sign-up sheet will go out closer to the date. Final submission and reproducibility package due **May 15**.

Announcement: Share Your Work

Two opportunities to get the word out about your research:

1. **Department blog post**
   - Write a short post for the economics department website about your capstone project
   - Great for your portfolio – shows employers you can communicate research to a broad audience
2. **Student ambassador interview**
   - The MSAE student ambassadors want to interview capstone students for promotional materials
   - Helps recruit future students and raises the profile of your cohort's work

Both are optional but encouraged. Talk to me after class if interested.

Check-in

What challenges are you facing in drafting your empirical strategy?

Reminder

Empirical Strategy Draft is due **tomorrow, Friday March 6** at 11:59pm. Today's session is designed to help you produce the figures and tables that support it.

EDA for Causal Inference

Learning Objectives

Today we focus on **looking at the data** before running models. Some of you have already started – that is great.

By the end of today, you will be able to:

1. Distinguish **descriptive EDA** from **causal EDA** – and know which checks matter for your identification strategy
2. Produce **key diagnostic plots** and a **summary statistics table** that support your empirical strategy draft
3. Apply a **reproducible figure workflow** – generate, save, and version-control publication-quality plots

EDA Is Not Just Description

Descriptive EDA

- What do the data look like?
- Distributions, outliers, missing values
- Summary statistics by group

Necessary, but not sufficient.

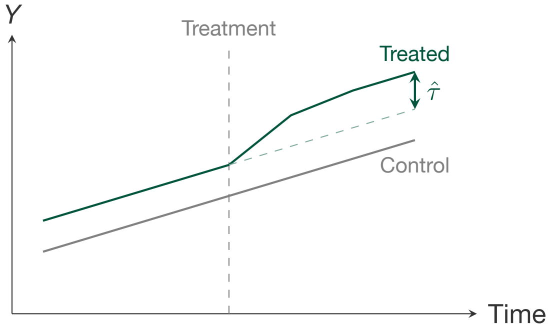
Causal EDA

- Do comparison groups look similar on **observables**?
- Does the data support your **identification assumptions**?
- Is the sample structured correctly for your method?

Directly tests your identification assumptions.

Bottom line: every figure in your paper should serve your causal argument, not just describe the data.

The Pre-Trends Plot



What to check:

1. Do groups track **together** before treatment?
2. Does a **gap** open after treatment?
3. Is the counterfactual (dashed) plausible?

Parallel pre-trends support (but do not prove) the parallel trends assumption.

Essential for DiD/SC/SDID.

EDA Checks by Method

Check	Why It Matters	Key Methods
Pre-trends	Groups move together before treatment	DiD, SC, SDID
Balance	Covariates similar across groups	All methods
First stage / density	Instrument strength (IV); smooth density at cutoff (RDD)	IV, RDD
Panel structure	All units observed in all periods	Panel FE, DiD
Outliers	No single unit driving the result	All methods

Run the checks relevant to *your* method before estimating.

Summary Statistics: What to Report

A good summary stats table shows:

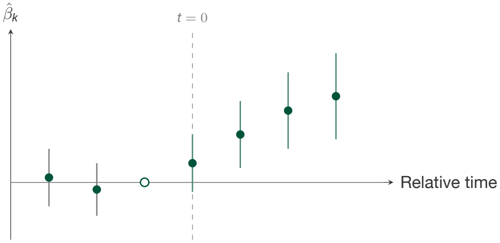
- Mean, SD, min, max for key variables
- Separate columns for comparison groups
- *N* for each group
- Difference + *p*-value for balance

This is Table 1 in most empirical papers.

```
library(dplyr)

df |>
  group_by(treated) |>
  summarize(
    n       = n(),
    mean_y  = mean(outcome, na.rm =
    TRUE),
    sd_y    = sd(outcome, na.rm = TRUE
    ),
    mean_x1 = mean(covariate1, na.rm =
     TRUE)
  )
```

The Event Study Plot



Reading an event study:

1. Pre-period coefficients near **zero** $\rightarrow$ parallel trends
2. Post-period **positive** $\rightarrow$ treatment effect
3. CIs exclude zero $\rightarrow$ significant

Hollow dot at $t = -1$ is the **reference period** (normalized to zero).

Dynamic version of pre-trends: tests the assumption ***period by period.***

EDA Checklist

Before running your main model, confirm you can check off each item:

- ✓ **Method-specific diagnostic** – pre-trends (DiD/SC), first stage (IV), density at cutoff (RDD)
- ✓ **Balance table** – covariates are similar across comparison groups (or you can explain why not)
- ✓ **Sample structure** – you know the units, periods, and any panel gaps or entry/exit patterns
- ✓ **Outlier check** – no single unit or period dominates the variation
- ✓ **Missing data audit** – you know where the gaps are and whether they are systematic

If any of these fail, fix the problem before estimating. Reviewers will ask about every one of these.

Early Figures Workshop

Workshop: Your First EDA Figure

20 minutes. Produce one EDA figure for your project.

Task A – Diagnostic plot for your method:

- DiD/SC: pre-trends plot (starter code on next slide)
- IV: first-stage scatter or binned plot of instrument vs. treatment
- RDD: outcome vs. running variable with local polynomial fit

Task B – Summary statistics (everyone should do this):

- Produce a summary table by comparison groups
- Identify which covariates are imbalanced

Use Claude Code: describe what you want, paste the starter code, and iterate.

Starter Code: Pre-Trends Plot

```
library(ggplot2)
library(dplyr)

# Compute group means by period
plot_df <- df |>
  group_by(year, treated) |>
  summarize(mean_y = mean(outcome, na.rm = TRUE),
            .groups = "drop") |>
  mutate(group = ifelse(treated == 1, "Treated", "Control"))

ggplot(plot_df, aes(x = year, y = mean_y,
                    color = group, linetype = group)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  geom_vline(xintercept = 2020, linetype = "dashed") +
  scale_color_manual(values = c("Control" = "gray50",
                                "Treated" = "#00543C")) +
  labs(x = "Year", y = "Mean Outcome", color = NULL,
       linetype = NULL) +
  theme_minimal(base_size = 14)
```

Figure Quality Standards

Common problems:

- Default gray ggplot2 background
- Tiny axis labels or legends
- No axis titles or unclear units
- Unlabeled treatment date
- Too many colors or groups

Fixes:

- Use `theme_minimal()` or `theme_bw()`
- Set `base_size = 14+` (slides need bigger fonts than papers)
- Label axes with units (e.g., "Employment rate (%)")
- Add `geom_vline()` with annotation
- Limit to 2–4 groups per plot

Rule of thumb: if you have to squint, the figure needs work. Every figure should be readable at slide size *and* in a printed paper.

Saving Figures Reproducibly

```
ggsave("output/figures/pre_trends.png",
       width = 8, height = 5, dpi = 300)
```

Best practices:

- Save to `output/figures/` – keep figures out of your code directory
- Use `.png` for slides, `.pdf` for papers (vector graphics)
- Set explicit `width`, `height`, and `dpi` so the figure looks the same everywhere
- Generate figures from a script, never save from the RStudio plot pane

If your figure is not generated by a script, it is not reproducible.

Workshop Share-out

2 volunteers, 3 minutes each.

Show your plot (or summary table) and tell us:

1. What are you plotting and why?
2. What does it tell you about your identification strategy?
3. Did anything surprise you?

Listeners: one suggestion for improving the figure.

Work Sprint

Sprint Goals

50 minutes. Choose the track that matches where you are:

Track	Goal
A: EDA	Produce 2–3 figures: diagnostic plot for your method, balance table, sample structure summary. Save to `output/figures/`.
B: Draft polish	Write or refine your empirical strategy section using today's EDA output as evidence. Include your DAG and key threats.

By the end of this sprint, you should have either new figures committed to Git or a substantially revised draft section.

I will circulate to help with both tracks.

Using Claude Code for EDA

Task	Prompt pattern
Pre-trends	"Plot mean [outcome] by year for treated vs. control. Add a vertical line at [year]."
First stage	"Regress [treatment] on [instrument]. Plot a binned scatter of the relationship."
RDD plot	"Plot [outcome] vs. [running var] with a local polynomial fit on each side of [cutoff]."
Balance table	"Compute means and SDs of [vars] by group. Add a column for the *p*-value of the difference."
Event study	"Estimate an event study with leads/lags using `fixest::feols()`. Plot coefficients with CIs."
Save figures	"Save all figures to output/figures/ at 300 dpi, 8x5 inches. Commit the script to Git."

Empirical Strategy Draft: What It Should Contain

Your draft (due tomorrow) should cover five components:

1. **Identification strategy** – what method, what variation, why it is plausibly exogenous
2. **Causal diagram** – your DAG with adjustment set identified (from Week 4)
3. **Main specification** – the regression equation you plan to estimate, with variables defined
4. **Key threats** – what could go wrong and how you will test for it (e.g., pre-trends, placebo, first-stage F, McCrary test)
5. **Preliminary evidence** – at least one figure or table from today's EDA

*This does not need to be polished – it needs to be **specific**. A clear draft gives me something concrete to give feedback on.*

Sprint Checkpoint

Before you stop, check:

1. Did you produce at least **one EDA figure** (diagnostic plot, balance table, or distribution)?
2. Does your figure support or challenge your **identification strategy**?
3. Is the figure generated by a **script** (not copy-pasted from the console)?
4. Did you **save** the figure to `output/figures/` with explicit dimensions?
5. Is your empirical strategy draft **started** (even if rough)?

If you answered “no” to #1, take 5 more minutes. Even a rough figure is better than none.

Key Takeaways

1. **EDA for causal inference** is not just description – every figure should test or support your identification assumptions
2. **Every method has a key diagnostic** – pre-trends (DiD), first stage (IV), density plots (RDD) – run yours before estimating
3. **Figures must be reproducible**: generated by a script, saved with explicit dimensions, version-controlled
4. **Start with the simplest plot**, then iterate – a rough figure today is worth more than a perfect figure next week

Wrap-Up & Next Week

Due tomorrow:

- **Empirical Strategy Draft – Friday March 6** at 11:59pm
- Include at least one figure or table from today's EDA

Next week (Week 7):

- Methods workshop: DiD, event studies, IV, panel FE, ML
- Hands-on estimation with your own data

Before next class:

- Submit your empirical strategy draft
- Push your EDA scripts and figures to GitHub
- Read the methods overview I will post (covers the estimators we will use in class)